

Network Topology

Following are the IPs allocated to individual systems in the network

1st Floor

- Reception – VLAN 80, Network: 192.168.8.0/24
- Store – VLAN 70, Network: 192.168.7.0/24
- Logistics – VLAN 60, Network: 192.168.6.0/24

2nd Floor

- Finance – VLAN 50, Network: 192.168.5.0/24
- HR – VLAN 40, Network: 192.168.4.0/24
- Sales/Marketing – VLAN 30, Network: 192.168.3.0/24

3rd Floor

- Admin – VLAN 20, Network: 192.168.2.0/24
- IT – VLAN 10, Network: 192.168.1.0/24

Step-by-Step process of configuration :

Routers : 2901 (3 routers , each on every floor)

Enable serial port on routers for connections

Added HWIC-2T

Cabel : serial DCE

Network between the routers are :

F1 - F2 : 10.10.10.0/30

F2 - F3 : 10.10.10.8/30

F3 - F1 : 10.10.10.4/30

Switch : 2960-24TT

Configuration of the components :

Router configurations

Router 1	Router 2	Router 3
Enable Conf t Int se0/2/0 No shutdown Int se0/2/1 No shutdown Int gig0/0 No shutdown	Enable Conf t Int se0/2/0 No shutdown Int se 0/2/1 No shutdown Int gig0/0 No shutdown Do write Int se0/2/1 Clock rate 64000 Do write	Enable Conf t Int se0/2/0 No shutdown Int gig0/0 No shutdown Int se0/2/0 Clock rate 64000 Int se 0/2/1 Clock rate 64000 Do write

Configuration of VLAN with Switches

Switch 1	Switch 2	Switch 3
Enable Conf t Int range fa0/2-3 Switchport mode access Switchport access vlan80 Enable Conf t Int range fa0/4-5 Switchport mode access Switchport access vlan70 Enable Conf t Int range fa0/6-8 Switchport mode access Switchport access vlan60 Do write	Enable Conf t Int range fa0/2-3 Switchport mode access Switchport access vlan50 Enable Conf t Int range fa0/4-5 Switchport mode access Switchport access vlan40 Enable Conf t Int range fa0/6-8 Switchport mode access Switchport access vlan30 Do write	Enable Conf t Int range fa0/2-3 Switchport mode access Switchport access vlan10 Enable Conf t Int range fa0/4-6 Switchport mode access Switchport access vlan20 Do write
Trunk Interface	Trunk Interface	Trunk Interface
En Conf t Int range f0/1 Switch port mode trunk Do write	En Conf t Int range f0/1 Switch port mode trunk Do write	En Conf t Int range f0/1 Switch port mode trunk Do write

Inter VLAN Configuration

Router 1	Router 2	Router 3
En Conf t Int gig0/0.80 Encapsulation dot1Q 80 Ip address 192.168.8.1 255.255.255.0 (Default gateway) Ex Int gig0/0.70 Encapsulation dot1Q 70 Ip address 192.168.7.1 255.255.255.0 (Default gateway) Ex Int gig0/0.60 Encapsulation dot1Q 60 Ip address 192.168.6.1 255.255.255.0 (default gateway) Do write Ex	En Conf t Int gig0/0.30 Encapsulation dot1Q 30 Ip address 192.168.3.1 255.255.255.0 (Default gateway) Ex Int gig0/0.40 Encapsulation dot1Q 40 Ip address 192.168.4.1 255.255.255.0 (Default gateway) Ex Int gig0/0.50 Encapsulation dot1Q 50 Ip address 192.168.5.1 255.255.255.0 (default gateway) Do write Ex	Int gig0/0.10 Encapsulation dot1Q 10 Ip address 192.168.1.1 255.255.255.0 (Default gateway) Ex Int gig0/0.20 Encapsulation dot1Q 20 Ip address 192.168.2.1 255.255.255.0 (default gateway) Do write Ex

Dhcp Configuration

Router 1	Router 2	Router 3
Enable Conf t Service dhcp Ip dhcp pool Reception Network 192.168.8.0 255.255.255.0 Default-router 192.168.8.1 dns -server 192.168.8.1 Ip dhcp pool Store Network 192.168.7.0 255.255.255.0 Default-router 192.168.7.1 dns -server 192.168.7.1 Ip dhcp pool Logistics Network 192.168.6.0 255.255.255.0 Default-router 192.168.6.1 dns -server 192.168.6.1 Do write Ex	Enable Conf t Service dhcp Ip dhcp pool finance Network 192.168.5.0 255.255.255.0 Default-router 192.168.5.1 dns -server 192.168.5.1 Ip dhcp pool HR Network 192.168.4.0 255.255.255.0 Default-router 192.168.4.1 dns -server 192.168.4.1 Ip dhcp pool Sales Network 192.168.3.0 255.255.255.0 Default-router 192.168.3.1 dns -server 192.168.3.1 Do write Ex	Ip dhcp pool IT Network 192.168.1.0 255.255.255.0 Default-router 192.168.1.1 dns -server 192.168.1.1 Ip dhcp pool Admin Network 192.168.2.0 255.255.255.0 Default-router 192.168.2.1 dns -server 192.168.2.1 Do write Ex

OSPF configuration

Router 1	Router 2	Router 3
Enable Conf t Router ospf 10 Network 10.10.10.4 255.255.255.252 area 0 Network 10.10.10.8 255.255.255.252 area 0 Network 192.168.8.0 255.255.255.0 area 0 Network 192.168.7.0 255.255.255.0 area 0 Network 192.168.6.0 255.255.255.0 area 0 Do write	En Conf t Router ospf 10 Network 10.10.10.0 255.255.255.252 area 0 Network 10.10.10.8 255.255.255.252 area 0 Network 192.168.3.0 255.255.255.0 area 0 Network 192.168.4.0 255.255.255.0 area 0 Network 192.168.5.0 255.255.255.0 area 0 Do write	Enable Conf t Router ospf 10 Network 10.10.10.0 255.255.255.252 area 0 Network 10.10.10.4 255.255.255.252 area 0 Network 192.168.1.0 255.255.255.0 area 0 Network 192.168.2.0 255.255.255.0 area 0 Do write

Now the device in the network can communicate with each other successfully

#Wireless network configuration

- For laptop we have add wireless module named as WPC300N

Access point configuration (Same process on all Access point on each Floor)

Access point F2

SSID:floor1

Password : floor@123

SSH Configuration

Router 1	Router 2	Router 3
En Conf t Hostname F1-router Ip domain-name adi Username adi password adi Crypto key generate rsa % How many bits in the module : 1024 Line vty 0 15 Login local Transport input ssh Do write Exit	En Conf t Hostname F2-router Ip domain-name adi Username adi password adi Crypto key generate rsa % How many bits in the module : 1024 Line vty 0 15 Login local Transport input ssh Do write Exit	En Conf t Hostname F3-router Ip domain-name adi Username adi password adi Crypto key generate rsa % How many bits in the module : 1024 Line vty 0 15 Login local Transport input ssh Do write Exit

-x-x-x- THE END -x-x-x-x